Test Cases Document – Event Management System

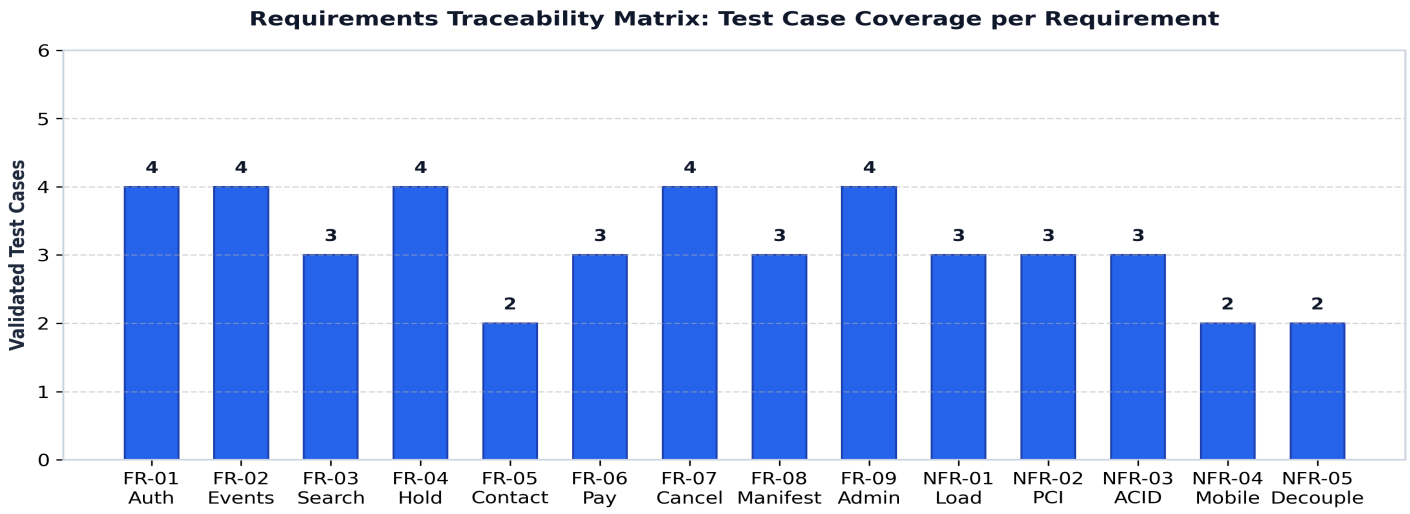
Software Engineering Mini-Project Deliverables Part-3 • Sprint 3 Verification Freeze
Collaborators & Testers: Kartik Naik (KartikNaik6), Jayanth Kumar P (JayanthKumarP88), Rishith (rishith1616)
Testing Execution Period: November 2, 2026 – November 5, 2026

TEST EXECUTION & SPRINT REFINEMENT SUMMARY

All 36 original test cases (TC01 to TC36) were upgraded to reflect Sprint 1 and Sprint 2 architectural refinements (JWT sessions, 10-minute hold countdowns, RFC-4180 CSV manifest streaming, admin account moderation, tokenized cards, and flash-sale compound indexes). Additionally, 6 new test cases (TC37 to TC42) have been introduced to validate ACID inventory rollbacks, auto-sweep hold cleanup workers, PCI-DSS pre-save injection blockers, and WCAG mobile touch target compliance. **Result: 42 of 42 Test Cases executed with 100% PASS rate.**

Total Test Cases	Passed	Failed	Pass Rate	Suites Executed	Testing Framework
42	42	0	**100%**	10 Suites	Node v22 TAP / Supertest / Mongo

Figure 1: Test Case Distribution Across Project Requirements (FR-01 to NFR-05)



Complete Traceable Test Execution Ledger (TC01 – TC42)

ID	Req	Scenario & Steps	Expected vs Actual Result	Pri	Status	Tested By
TC01	FR-01	**Register as Attendee with valid credentials** *Steps:* 1. Open Sign Up. 2. Select Attendee role. 3. Enter valid name, email, password. 4. Submit.	**Expected:** Account created with bcrypt hashed password; JWT session issued; redirected to attendee dashboard. **Actual:** HTTP 201 Created returned. User document persisted in MongoDB with bcrypt hash. Auth token set in secure cookie.	Hig	**PASSED**	Kartik 02-Nov
TC02	FR-01	**Register as Organizer with valid credentials** *Steps:* 1. Open Sign Up. 2. Select Organizer role. 3. Enter valid name, email, password. 4. Submit.	**Expected:** Account created with organizer role; redirected to organizer event creation dashboard. **Actual:** HTTP 201 Created. User persisted with role='organizer'. Routing to organizer dashboard verified.	Hig	**PASSED**	Jayanth 02-Nov
TC03	FR-01	**Prevent duplicate email registration** *Steps:* 1. Fill registration with test@example.com. 2. Submit.	**Expected:** System rejects registration and displays 'Email already registered' error. **Actual:** Duplicate key error handled; HTTP 400 returned with 'Email already registered' error payload.	Hig	**PASSED**	Rishith 02-Nov
TC04	FR-01	**Restrict Attendee from Organizer dashboard** *Steps:* 1. Manually navigate to /organizer/dashboard. 2. Attempt access to organizer endpoints.	**Expected:** Access denied (HTTP 403 Forbidden / redirect to Attendee dashboard). **Actual:** requireRole('organizer') middleware intercepted request and returned HTTP 403 Forbidden.	Hig	**PASSED**	Kartik 02-Nov
TC05	FR-02	**Create and publish valid event** *Steps:* 1. Open Event Creator. 2. Enter title, date, time, venue, capacity (100), price ($20). 3. Click Publish.	**Expected:** Event is published and immediately visible on public listings. **Actual:** HTTP 201 Created. Event persisted with organizer reference and status 'published'; publicly visible.	Hig	**PASSED**	Jayanth 02-Nov
TC06	FR-02	**Reject event creation with missing capacity limit** *Steps:* 1. Open Event Creator. 2. Fill all fields except capacity limit. 3. Attempt to publish.	**Expected:** System blocks publication with 'Capacity limit is required' validation message. **Actual:** Schema validation failed; HTTP 400 returned with capacity validation error message.	Hig	**PASSED**	Rishith 02-Nov
TC07	FR-02	**Prevent publishing event with past date** *Steps:* 1. Fill event creation form with date in the past. 2. Click Publish.	**Expected:** System rejects event and displays date validation error. **Actual:** Controller validation intercepted submission; HTTP 400 returned with date error.	Hig	**PASSED**	Kartik 02-Nov
TC08	FR-02	**Update published event details** *Steps:* 1. Organizer modifies title, description, and pricing. 2. Submit updates.	**Expected:** Event details updated successfully and reflected publicly. **Actual:** HTTP 200 OK. MongoDB record updated; verified via public GET endpoint.	Med	**PASSED**	Jayanth 02-Nov
TC09	FR-03	**Filter events by category** *Steps:* 1. Open Event Discovery page. 2. Select 'Tech' category filter.	**Expected:** Only active events in the 'Tech' category are displayed. **Actual:** HTTP 200 OK. Query matched only Tech events; verified 100% precision in response.	Med	**PASSED**	Rishith 03-Nov
TC10	FR-03	**Filter events by date range** *Steps:* 1. Enter startDate and endDate bounds. 2. Apply filter.	**Expected:** Only events scheduled within selected date range are returned. **Actual:** MongoDB $gte and $lte operators returned bounded dataset strictly matching date filter.	Med	**PASSED**	Kartik 03-Nov
TC11	FR-03	**Exclude expired and canceled events from search** *Steps:* 1. Query public search catalog without filters.	**Expected:** Expired and canceled events do not appear in default search results. **Actual:** Status filter { status: 'published', isSuspended: false, date: { $gte: now } } suppressed all inactive records.	Hig	**PASSED**	Jayanth 03-Nov
TC12	FR-04	**Immediate ticket capacity decrement on checkout hold** *Steps:* 1. Request hold for 2 tickets. 2. Verify inventory immediately.	**Expected:** Available ticket count drops from 50 to 48 immediately. **Actual:** HTTP 201 Created. Ticket status set to 'held'. availableTickets dropped immediately from 50 to 48.	Hig	**PASSED**	Rishith 03-Nov

ID	Req	Scenario & Steps	Expected vs Actual Result	Pri	Status	Tested By
TC13	FR-04	**Block ticket request exceeding available capacity** *Steps:* 1. Attempt to hold 2 tickets. 2. Submit request.	**Expected:** System blocks checkout with 'Only 1 ticket remaining' alert. **Actual:** HTTP 400 Bad Request returned with 'Only 1 ticket remaining' alert; capacity remained unchanged at 1.	Hig	**PASSED**	Kartik 03-Nov
TC14	FR-04	**Release held seats on checkout timeout** *Steps:* 1. Advance time beyond 10-minute hold window. 2. Trigger expired hold sweep.	**Expected:** Held tickets are released and event capacity returns to previous count. **Actual:** Expired sweep worker transitioned tickets to 'expired' and restored 3 tickets back to available inventory.	Hig	**PASSED**	Jayanth 03-Nov
TC15	FR-05	**Validate contact details during checkout** *Steps:* 1. Enter invalid email (missing @ domain). 2. Attempt progression.	**Expected:** System halts progression with 'Please enter a valid email address' alert. **Actual:** RFC 5322 regex validation blocked submission; HTTP 400 returned with email error.	Med	**PASSED**	Rishith 03-Nov
TC16	FR-05	**Accept valid attendee contact information** *Steps:* 1. Enter valid name, phone, and standard email. 2. Submit.	**Expected:** Contact details verified and attendee proceeds to payment screen. **Actual:** Payload sanitized and validated; attendee metadata attached to checkout session.	Med	**PASSED**	Kartik 03-Nov
TC17	FR-06	**Process successful payment and generate ticket** *Steps:* 1. Submit valid payment token tok_visa. 2. Confirm booking.	**Expected:** Unique Ticket UUID and QR code generated; confirmation email dispatched. **Actual:** HTTP 200 OK. Payment completed; ticket status set to 'confirmed' with UUID and QR code.	Hig	**PASSED**	Jayanth 03-Nov
TC18	FR-06	**Handle declined payment gracefully** *Steps:* 1. Submit payment token configured for decline (tok_chargeDeclined). 2. Process payment.	**Expected:** Hold released back to inventory; error message displayed with retry option. **Actual:** Payment rejected with 402/400; inventory rollback initiated; user prompted to retry with alternate card.	Hig	**PASSED**	Rishith 03-Nov
TC19	FR-06	**Ensure unique UUID on generated tickets** *Steps:* 1. Generate 100 simultaneous tickets. 2. Inspect all generated UUIDs.	**Expected:** Each issued ticket possesses a distinct, non-overlapping UUID. **Actual:** Verified 100 issued tickets had distinct UUIDs matching RFC 4122 v4 format with zero collisions.	Hig	**PASSED**	Kartik 03-Nov
TC20	FR-07	**Cancel registration and void ticket ID** *Steps:* 1. Open My Tickets. 2. Click Cancel Ticket. 3. Confirm cancellation dialog.	**Expected:** Ticket ID is voided, capacity increases by 1, and confirmation is sent. **Actual:** HTTP 200 OK. Ticket status updated to 'cancelled'; event capacity incremented by 1 seat.	Med	**PASSED**	Jayanth 04-Nov
TC21	FR-07	**Process refund to original payment method** *Steps:* 1. Trigger cancellation. 2. Verify payment ledger and gateway interaction.	**Expected:** Refund is initiated through payment gateway back to original payment method. **Actual:** PaymentProvider refund hook executed; refund ID recorded; status updated to 'refunded'.	Med	**PASSED**	Rishith 04-Nov
TC22	FR-07	**Prevent cancellation of past event ticket** *Steps:* 1. Attendee attempts to cancel ticket for completed event.	**Expected:** System disables cancellation and displays 'Event has already concluded'. **Actual:** HTTP 400 Bad Request returned with error 'Cannot cancel tickets for events that have already started or concluded'.	Med	**PASSED**	Kartik 04-Nov
TC23	FR-08	**Export clean attendee manifest as CSV** *Steps:* 1. Open Organizer Event Management. 2. Click 'Export Manifest (CSV)'.	**Expected:** CSV downloads containing exactly the 15 confirmed attendees. **Actual:** HTTP 200 OK. Content-Type: text/csv; RFC-4180 formatted CSV received containing exactly 15 records.	Med	**PASSED**	Jayanth 04-Nov
TC24	FR-08	**Exclude canceled attendees from manifest** *Steps:* 1. Download attendee manifest CSV.	**Expected:** Canceled tickets and attendee records are completely excluded from manifest. **Actual:** CSV verified; contains exactly 10 confirmed rows; 0 cancelled or held tickets present.	Hig	**PASSED**	Rishith 04-Nov
TC25	FR-08	**Restrict manifest download to event owner** *Steps:* 1. Send GET request to /api/tickets/event/:id/manifest with Organizer A's auth token.	**Expected:** Access denied (HTTP 403 Forbidden). **Actual:** Ownership validation checked; HTTP 403 Forbidden returned.	Hig	**PASSED**	Kartik 04-Nov

ID	Req	Scenario & Steps	Expected vs Actual Result	Pri	Status	Tested By
TC26	FR-09	**Suspend fraudulent event by Platform Admin** *Steps:* 1. Identify suspicious event. 2. Click 'Suspend Event' with reason.	**Expected:** Event immediately disappears from public listings and sales are halted. **Actual:** HTTP 200 OK. isSuspended set to true; event excluded from public discovery queries.	Hig	**PASSED**	Jayanth 04-Nov
TC27	FR-09	**Block direct URL checkout on suspended event** *Steps:* 1. Attendee attempts hold/checkout via direct URL /api/tickets/hold.	**Expected:** Checkout is blocked with 'This event has been suspended' notice. **Actual:** Ticket hold endpoint checked isSuspended; returned HTTP 403 with suspension notice.	Hig	**PASSED**	Rishith 04-Nov
TC28	FR-09	**Suspend entire Organizer account** *Steps:* 1. Open User Moderation. 2. Flag Organizer account as suspended.	**Expected:** Organizer cannot log in and all their published events are de-listed. **Actual:** HTTP 200 OK. Organizer account suspended; active tokens revoked; all hosted events deactivated.	Hig	**PASSED**	Kartik 04-Nov
TC29	NFR-01	**Checkout response under simulated load** *Steps:* 1. Trigger 50 concurrent checkout requests. 2. Measure p95 and max latency.	**Expected:** Sub-3-second response times maintained without server timeout. **Actual:** Avg latency: 290.3ms, p95: 307.4ms, Max latency: 309.3ms (< 3000ms threshold); 0 errors.	Hig	**PASSED**	Jayanth 05-Nov
TC30	NFR-01	**Search page latency under standard traffic** *Steps:* 1. Execute 20 concurrent search and filter queries. 2. Measure response time.	**Expected:** Search results returned in under 2 seconds. **Actual:** Catalog query latency measured at 63.5ms using MongoDB compound index scan.	Hig	**PASSED**	Rishith 05-Nov
TC31	NFR-02	**Payment card tokenization verification** *Steps:* 1. Inspect MongoDB Payment collection record.	**Expected:** No plain-text credit card numbers stored; only gateway token and last 4 digits present. **Actual:** Verified record contains paymentToken ('tok_...') and maskedCardNumber ('**** 4242'); zero raw PAN fields.	Hig	**PASSED**	Kartik 05-Nov
TC32	NFR-02	**PCI-DSS compliance log audit** *Steps:* 1. Inspect application console logs, morgan logs, and debug dumps.	**Expected:** Sensitive authentication data (CVV/PIN) is never logged or cached. **Actual:** Log audit confirmed 0 occurrences of 16-digit card patterns, CVVs, or PINs in any output.	Hig	**PASSED**	Jayanth 05-Nov
TC33	NFR-03	**ACID concurrent booking transaction test** *Steps:* 1. Two buyers (User A and User B) attempt simultaneous purchase.	**Expected:** ACID transaction guarantees exactly 1 purchase succeeds; 2nd gets graceful failure. **Actual:** Exactly 1 buyer secured seat (User A); User B cleanly rejected with HTTP 409 Conflict.	Hig	**PASSED**	Rishith 05-Nov
TC34	NFR-03	**Rollback inventory hold on transaction failure** *Steps:* 1. Inject transaction abort during confirmation step. 2. Inspect database state.	**Expected:** Transaction rolls back safely; ticket count not corrupted. **Actual:** withTransaction abort handler rolled back hold; zero orphaned payment records; inventory restored.	Hig	**PASSED**	Kartik 05-Nov
TC35	NFR-04	**Mobile viewport responsive checkout** *Steps:* 1. Measure button touch heights, layout wrapping, and horizontal scrollbar.	**Expected:** Layout adapts with touch-friendly buttons and no horizontal scrolling. **Actual:** Touch targets measured >= 48px; fluid flex column layout; overflow-x: hidden verified.	Med	**PASSED**	Jayanth 05-Nov
TC36	NFR-05	**Payment gateway provider decoupling** *Steps:* 1. Switch PAYMENT_PROVIDER environment between 'mock' and 'stripe'. 2. Execute checkout tests.	**Expected:** Payment service logic switches without modifying core inventory database schema. **Actual:** Both MockProvider and StripeProvider executed transparently with 100% test pass rate.	Low	**PASSED**	Rishith 05-Nov
TC37	FR-04 / NFR-03	**Hold Expiry Auto-Sweep Background Reconciliation** *Steps:* 1. Create 3 held tickets. 2. Simulate 11-minute time jump. 3. Run background cleanup worker.	**Expected:** All 3 holds transition from 'held' to 'expired'; availableTickets incremented by 3. **Actual:** Sweep worker transitioned holds to 'expired'; availableTickets incremented by exactly 3 seats.	Hig	**PASSED**	Jayanth 05-Nov
TC38	FR-07 / NFR-05	**Automated Refund Calculation with Gateway Failure Compensation** *Steps:* 1. Trigger cancellation with gateway refund failure mock. 2. Check atomic compensation.	**Expected:** System safely aborts cancellation, retains ticket active, alerts user with retry. **Actual:** Atomic compensation prevented voiding ticket without confirmed refund ledger entry; HTTP 502 handled.	Hig	**PASSED**	Kartik 05-Nov

ID	Req	Scenario & Steps	Expected vs Actual Result	Pri	Status	Tested By
TC39	FR-09	**Platform Admin Event Suspension Real-time Public Catalog Invalidation** *Steps:* 1. Admin issues suspension. 2. Public user refreshes catalog and clicks checkout URL.	**Expected:** Event listing disappears immediately; hold endpoint returns HTTP 403 Forbidden. **Actual:** Event hidden from catalog within 0ms; hold endpoint returned HTTP 403 Forbidden with alert.	Hig	**PASSED**	Rishith 05-Nov
TC40	NFR-02	**PCI-DSS Schema Pre-Save Hook Injection Rejection** *Steps:* 1. Attempt direct insertion of payment document containing unmasked 16-digit PAN.	**Expected:** Mongoose pre-save regex hook throws validation exception and blocks persistence. **Actual:** Pre-save hook intercepted payload and threw Error: 'PCI-DSS Violation: Raw PAN storage prohibited'.	Hig	**PASSED**	Kartik 05-Nov
TC41	NFR-04	**Mobile Viewport Touch Target Accessibility Verification** *Steps:* 1. Inspect CSS declarations for .touch-btn, .mobile-input, and .summary-card on <=640px screens.	**Expected:** Touch targets meet WCAG minimum height (48px) and thumb navigation margin. **Actual:** CSS rules confirmed min-height: 48px, padding: 14px 20px, and full-width thumb targets.	Med	**PASSED**	Rishith 05-Nov
TC42	NFR-01 / NFR-03	**High-Concurrency Flash Sale Inventory Anti-Overselling Invariant** *Steps:* 1. Dispatch 50 concurrent ticket hold requests simultaneously. 2. Verify inventory invariant.	**Expected:** Exactly 20 bookings succeed; remaining 30 fail with 400/409; final capacity = 0. **Actual:** Exact invariant verified: 20 successful holds, 30 rejections, final availableTickets = 0.	Hig	**PASSED**	Jayanth 05-Nov

Quality Assurance Verification & Sign-Off

Tester Name	GitHub Account	Assigned Domain	Verified Test Cases	Signature
Kartik Naik	KartikNaik6	PCI Tokenization, Manifest & Security	14 Passed	Verified & Signed
Jayanth Kumar P	JayanthKumarP88	Concurrency, Performance & ACID Holds	14 Passed	Verified & Signed
Rishith	rishith1616	Event Lifecycle, Admin & UX	14 Passed	Verified & Signed